

Aerchain Assignment | Kaustubh Rege

Purpose:

Improve the efficiency of the purchase process from creation of a RFx to vendor selection.

Business Outcome:

Current process: Total days to finalise vendor: 3 weeks (As per brief)

Using Procurement assistant: 1 week(vendor response deadline 5 days)

2 weeks reduction in vendor finalisation

Other outcomes:

- Reduced time in making a RFx with consistent format across employees
- Clear comparison between vendor quotes
- Automatic filtering based on quality & compliances
- Interactive analysis using assistant with clear answers

What I built

I have divided the solution into 4 independent features, each running independently,

1. **RFx Creation:** Conversational co-pilot that drafts a complete RFx (scope, line items, questionnaire, commercial terms) via a forced Claude tool-call, then sends it (stubbed — logged, not real SMTP). Currently takes in only text, but should be multi modal in iterative versions.
2. **Ingest & Extract:** Converts any vendor file format into text or an image, then a forced Claude tool-call extracts structured line items, category-level pricing rules, buried document-level conditions, and questionnaire answers as schema-valid JSON.
3. **Normalize:** Deterministic Python that converts every vendor's price to one currency and unit (INR/piece), applies footnote-style adjustments, and resolves category rules into implied per-SKU prices. LLM used only for
4. **AI Analyst:** Conversational agent which answers free text questions from the buyer.
5. **Streamlit UI:** Ties it all together with tables and row level drill down, alongside conversational interfaces.

What I deliberately left out

- Real email/SMTP, vendor responses are simulated files, not an inbox integration.
- Live FX rates: USD→INR uses a documented constant (83), not a real-time rate lookup.
- OCR fallback for scanned (non-text-layer) PDFs — only handled via the image/vision path.

Edge Cases Covered

6% discount buried in a PDF footnote	Caught as a document-level adjustment, applied deterministically, every affected line flagged
Prices quoted in USD, no table (prose)	Extracted per-line despite no table; every line flagged currency-not-INR
Rates per-100-pieces, not per-piece	Converted ÷100 with an explicit flag on every converted cell
Only 27 of 30 SKUs quoted (photo)	3 missing SKUs shown as blank + “NOT QUOTED”, never guessed
Category rules instead of per-SKU prices (email)	Captured as rules, resolved to implied per-SKU prices by a separate labeled LLM step